

Databases and SQL in Simple Words

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Tables, keys and the four SQL commands every beginner needs.

Tables, rows and keys

A database stores data in tables. A table has columns (the fields) and rows (the records). A primary key is a column that identifies each row in a unique way, like a roll number.

A foreign key is a column that points to the primary key of another table. This is how tables are connected.

Four commands you use every day

SELECT reads data, INSERT adds a row, UPDATE changes a row and DELETE removes a row.

```
SELECT title, author FROM books WHERE category_id = 1;  
INSERT INTO categories (name) VALUES ('Science');
```

Always use WHERE with UPDATE and DELETE, or you will change every row.

Why we split data into tables

Imagine writing the category name inside every book row. If the name changes, you must fix hundreds of rows. Instead, keep categories in one table and connect books to it with a foreign key. This idea is called normalization, and it removes repeated data.